

**NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2 PRELIMINARY EXAMINATION
Higher 2**

MATHEMATICS

9758/01

Paper 1

24 August 2018

3 hours

Additional Materials: Answer Paper
 List of Formulae (MF26)
 Cover Sheet

READ THESE INSTRUCTIONS FIRST

Write your name, registration number, subject tutorial group, on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of **8** printed pages.



National Junior College

[Turn over

- 1 The n^{th} term of a sequence is given by $u_n = n!(n-2)$, for all positive integers n where $n \geq 2$.

Show that

$$u_n - u_{n-1} = (n-1)!(an^2 + bn + c)$$

for some real constants a , b and c to be determined. [2]

Hence find

$$\sum_{n=3}^{N+1} [(n-1)!(2n^2 - 6n + 6)]. \quad [3]$$

- 2 Referred to the origin O , the points A and B are such that $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. The point C is on AB produced such that $AB : BC = 1 : k$, where k is a real constant.

Given that $OA = 2OC$, $\angle AOB = \frac{\pi}{2}$, and $\angle AOC = \frac{3\pi}{4}$, find the exact value of k , showing your working clearly. [5]

- 3 By means of the substitution $u = \sin \theta$, find

$$\int \frac{2 \cos \theta - 3 \sin 2\theta}{1 + \sin^2 \theta} d\theta. \quad [5]$$

- 4 The curve C has equation $9x^2 - y^2 + 3xy - 5 = 0$.

- (i) Find $\frac{dy}{dx}$ in terms of x and y . Hence, explain why C has no turning points. [4]
- (ii) Find the equations of the tangent and the normal to C at the point $P(1, -1)$. [2]
- (iii) The tangent and normal to C at the point $P(1, -1)$ meets the y -axis at the points Q and R respectively. State the area of triangle PQR . [1]

- 5 Given that a is a positive constant. A curve C_1 has parametric equations

$$x = \frac{a}{t}, \quad y = 1 + t.$$

Sketch C_1 , labelling the coordinates of the point(s) where the curve crosses the x - and y -axes, and the equations of the asymptote(s) in terms of a , if any. [2]

Another curve C_2 has equation $y = \sqrt{1 + \frac{x^2}{a^2}}$.

- (i) Show algebraically that the y -coordinates of the point(s) of intersection of C_1 and C_2 satisfies the equation $(y-1)^2(y^2-1)-1=0$. [2]
- (ii) Sketch C_2 on the same diagram as C_1 , labelling the coordinates of the point(s) where the curve crosses the x - and y -axes, and the equations of the asymptotes in terms of a , if any.

Find the coordinates of point(s) of intersections of C_1 and C_2 and label the coordinates in this diagram, leaving the answers correct to 3 significant figures, in terms of a . [4]

- 6 (a) The function f is defined by

$$f : x \mapsto 2x^2 - \lambda x - 3, \text{ where } x \in \mathbb{R}, \frac{7}{4} < x < 5,$$

where λ is a real constant. Find the set of possible values of λ such that f^{-1} exists. [2]

- (b) The function g is defined by

$$g(x) = \begin{cases} (x-1)^2, & \text{for } 0 \leq x \leq 1, \\ 2\log_2 x, & \text{for } 1 < x \leq 2. \end{cases}$$

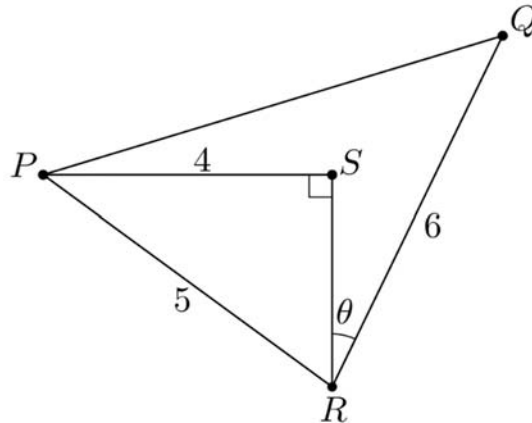
- (i) Sketch the graph of $y = g(x)$, labelling clearly the coordinates of the end-points and the points where the curve crosses the x -axis, if any. [2]
- (ii) Hence solve the inequality $1 < g(x) \leq 2$ exactly. [2]
- (iii) Given that g^{-1} exists, define g^{-1} in a similar form as g . [3]

- 7 (a) Given that $y + 2 = (x + 1)^{\ln(x+1)}$, where $x > -1$, show that

$$(x + 1) \frac{dy}{dx} = 2(y + 2) \ln(x + 1). \quad [2]$$

By repeated differentiation of the above result, find the Maclaurin series of y up to and including the term in x^2 . [3]

(b)



In the diagram above, $QR = 6$, $PS = 4$, $PR = 5$, $\angle PSR = \frac{\pi}{2}$ and $\angle QRS = \theta$ radians.

- (i) Show that $PQ = (61 - 36 \cos \theta + 48 \sin \theta)^{\frac{1}{2}}$. [2]
- (ii) Given that θ is a sufficiently small angle, show that

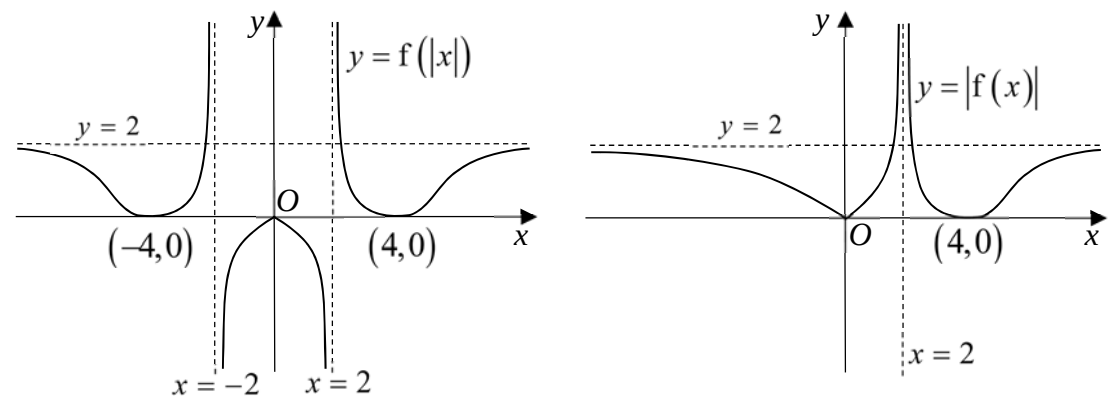
$$PQ \approx 5 + p\theta + q\theta^2$$

for some rational constants p and q to be determined exactly. [4]

- 8 (a) Describe a sequence of exactly three transformations that will transform the curve with equation $y = \frac{1}{x-3}$ onto the curve with equation $y = \frac{2x+a}{3-x}$, where a is a positive constant. [3]

Given instead that $a < -6$, use a non-graphical method to determine the range of values of x where the graph of $y = \frac{2x+a}{3-x}$ is concave upwards. [2]

- (b) The diagrams below show the graphs of $y = f(|x|)$ and $y = |f(x)|$, where the equations of the asymptotes and the coordinates of the turning points are given. The gradient of the graph with equation $y = f(x)$ at the origin is $-\frac{3}{4}$.



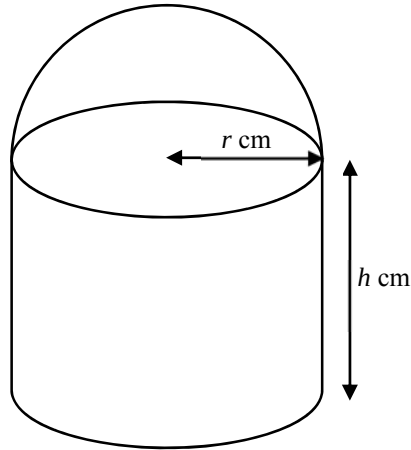
On separate diagrams, sketch the graphs of

- (i) $y = f(x)$, [2]
 (ii) $y = \frac{1}{f(x)}$, [3]
 (iii) $y = f'(x)$, [3]

including the coordinates of the points where the graphs cross the x - and y - axes and the equations of the asymptotes, if any.

- 9 [It is given that a sphere of radius r has surface area $4\pi r^2$ and volume $\frac{4}{3}\pi r^3$.]

A manufacturer produces closed hollow cans of fixed volume $k \text{ cm}^3$ as shown in the diagram below. The top part is a hemisphere made of tin. The bottom part is a cylinder made of aluminium of cross-sectional radius $r \text{ cm}$ and height $h \text{ cm}$. There is no material between the cylinder and the hemisphere so that any fluid can move freely within the container.



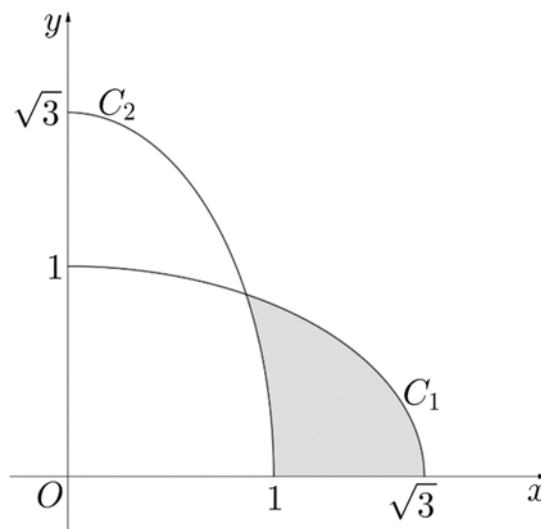
- (i) If tin costs 4 cents per cm^2 and aluminium costs 6 cents per cm^2 , use differentiation to find the values of r and h such that the total cost of producing the cans is minimised, giving your answers in terms of k . Simplify your answers. [7]
- (ii) At the beginning of an experiment, a similar-shaped can of dimensions $r = 4$ and $h = 10$, is filled to its capacity with water. Due to a hole at its base, water is leaking at a constant rate of $2 \text{ cm}^3 \text{ s}^{-1}$ when the can is standing upright. Find the exact rate at which the height of the water is decreasing 80 seconds after the start of the experiment. [5]

- 10 (i) Show that for any real constant p , $\frac{p^2 x^2}{\sqrt{1-p^2 x^2}} = \frac{1}{\sqrt{1-p^2 x^2}} - \sqrt{1-p^2 x^2}$.

Hence, or otherwise, prove that for any constant n such that $0 < n < \frac{1}{p}$,

$$\int_n^{\frac{1}{p}} \sqrt{1-p^2 x^2} \, dx = \frac{\pi}{4p} - \frac{1}{2p} \sin^{-1}(pn) - \frac{1}{2} n \sqrt{1-p^2 n^2}. \quad [6]$$

- (ii) The curves C_1 and C_2 are ellipses with the origin O as their common centre. It is given that the points $(0, 1)$ and $(\sqrt{3}, 0)$ are vertices of C_1 and the points $(0, \sqrt{3})$ and $(1, 0)$ are vertices of C_2 . The diagram below shows the parts of the two curves in the first quadrant.



Use the result in part (i) to find the area of the shaded region exactly, giving your answer in the form $\frac{\pi\sqrt{3}}{m}$, where m is an integer constant to be determined. [6]

[Question 11 is printed on the next page.]

- 11** A swimming pool contains 375000 litres of pure water. Water containing s milligrams of free chlorine per litre flows into the pool at a rate of 10 litres per minute. The pool is also draining at a rate of 10 litres per minute, such that the volume of the water in the pool remains constant.

- (i) Show that the rate at which the mass of the free chlorine, x grams, is changing in the pool over time, t minutes, can be modelled by the differential equation

$$\frac{dx}{dt} = \frac{375s - x}{37500},$$

State a necessary assumption for the above model to be valid. [3]

- (ii) Given that no free chlorine is present in the pool initially, find x in terms of t and s . [4]

- (iii) Find, in terms of s , the mass of free chlorine present in the pool after one hour. [1]

- (iv) Sketch the graph of x against t which is relevant to the context, labelling the point(s) where the curve crosses the axes and the equation(s) of the asymptote(s). Hence determine the mass of free chlorine present in the pool after a long period in terms of s . [3]

To prevent health complications, the recommended safe level of free chlorine to be used is between 375 grams and 1125 grams in a pool which contains 375 000 litres of water. Find the range of values of s that should be used. [2]

– END OF PAPER –